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TECHNICAL SUPPLEMENT TO NEURAL BELLMAN OPERATORS

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This supplement gives complete proofs of own-future construction, reference-centered risk qualification, continuous investment and decision-scale precision. It adds the action-support and prospective-selection arguments, nonuniform-law enclosures, explicitly quantified randomized information bounds, and exact economic decision separation. Complete prior expectation, signed-risk and uniform-information proofs remain in the Development Appendix.

1. PROOFS OF THE INHERITED NBO RESULTS

These proofs retain the original statements and arguments. Their placement in the supplement changes the reading order, not their hypotheses.

1.1. *Proof of the result 1*

PROOF: Backward comparison and cash invariance give

$$V_t^* \geq f_t - \sum_{k=t}^{T-1} B_{t,k} u_k - B_{t,T} u_T, \quad J_t^\pi \leq f_t + \sum_{k=t}^{T-1} B_{t,k} v_k - B_{t,T} \ell_T,$$

where $B_{t,k} = \prod_{j=t}^{k-1} \beta_j$. The terminal claim is the stipulated enclosure; applying the Bellman or fixed-policy operator proves each inductive step. Subtraction gives (3). A shift $f_t \mapsto f_t + c_t^0$ adds $c_t^0 - \beta_t c_{t+1}^0$ to u_t and subtracts it from v_t. The terminal width is unchanged. *Q.E.D.*

1.2. *Proof of the result 2*

PROOF: The selected node label lies between the node Bellman infimum minus e_t and that infimum plus $D_t k_t + e_t$. The Lipschitz property of the infimum implies that every cone lies above it minus e_t. At a nearest node its cone exceeds the infimum by

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at most $D_t k_t + e_t + 2L_t h_t$. At the attaining original index, the two feasible pairs and (4) give

$$Q_t(f_{t+1}; x, \Gamma_{t,i}(x)) \le y_{t,i} + e_t + L_t \|x - x_{t,i}\|_1 + D_t \zeta_t.$$

This proves (6). The terminal residual belongs to $[-e_T, e_T + 2L_g h_T]$; Proposition 1 gives the loss bound. A finite minimum of equally Lipschitz cones is equally Lipschitz. The identities $|z| = \mathrm{ReLU}(z) + \mathrm{ReLU}(-z)$ and $\min(u, v) = u - \mathrm{ReLU}(u - v)$ give the exact network. Finite lexicographic comparison and Borel repair give measurability. Effective refinement of the finitely many positively weighted terms yields the termination assertion. *Q.E.D.*

1.3. *Proof of the result 3*

PROOF: The triangle inequality bounds $f(x)$ above by each corner expression. Choose the lexicographically first attaining site $i_*(x)$. In every coordinate choose the cell endpoint between x_j and $x_{i_*,j}$. At this corner distances add exactly. An owner of strictly smaller transformed value would improve the objective at x; a tied smaller owner would contradict the original tie rule. Thus this corner has owner $i_*(x)$, proving both identities. Along a coordinate line, forward and backward relaxations respectively minimize over sites on either side, always comparing value–original-index pairs. Iterating over coordinates gives the separable distance transform and its stated work. Rational arithmetic and unchanged repair complete the claim. *Q.E.D.*

1.4. *Proof of the result A.60*

PROOF: The state cost has gradient infinity norm at most $8/d$, Hessian norm at most $22/d$, and diagonal second derivatives at most $5/d + 16/d^2$. For the terminal cost the corresponding bounds are $10/d$, $26/d$, and $9/d + 16/d^2$. They follow by differentiating the quadratic terms and the continuously differentiable squared shortage. The bound on the cycle Laplacian is four. The shortage Hessian, where it exists, is $16\mathbf{1}\mathbf{1}^\top/d^2$; the same second-difference bound holds across its kink.

For feasible (x, a) and (y, b), put $w = x - y$. The difference between the transition at their convex combination and the corresponding combination of transitions is $\theta(1-\theta) w_j w_{j+1}/16$ in coordinate j. Convexity and the G_{t+1} Lipschitz bound

therefore give a convexity defect of at most

$$\frac{\beta G_{t+1}}{16}\theta(1-\theta)\sum_j |w_j w_{j+1}| \le \frac{\beta G_{t+1}}{16}\theta(1-\theta)\|w\|_2^2.$$

The state cost supplies curvature $4/d$, while the two-control action cost has Hessian eigenvalues at least $7/4$. Thus the state curvature of Q_t^* is at least $4/d - \beta G_{t+1}/8$, which is at least $107/(128d)$ when $G_{t+1} \le 27/d$. Partial minimization over the affine capacity graph preserves the state convexity inequality. The minimizing action is unique, including at a binding capacity constraint.

To propagate Lipschitz continuity, write $a = c(x)u$, where u lies in the fixed unit simplex. Transfer the same u from x to y. Each action-cost partial derivative has absolute value at most $13/16$, and $\sum_j u_j \le 1$. The action-cost transfer contributes at most $13\|x-y\|_1/(128d)$. The transition Jacobian at a fixed action has column sums at most $11/16$; the capacity transfer adds at most $3/64$. Comparing a minimizing normalized action in both directions gives the stated recursion for G_t. Its right side maps $27/d$ below itself.

For a fixed u, the transition as a function of x has Jacobian norm at most $11/16 + 1/16 = 3/4$. Its second-order composition term is bounded by $G_{t+1}/8$. The action-cost Hessian has norm at most $21/4$, and the capacity gradient has squared norm $1/(64d)$. These facts give the recursion for M_t. The infimum over u preserves this common semiconcavity bound: after subtracting the same quadratic, each fixed-u function is concave, and an infimum of concave functions is concave. The right side maps $54/d$ below itself. Convexity together with semiconcavity yields the claimed differentiability and gradient regularity; this conclusion does not require a differentiable optimizing-action map.

Along one state coordinate, the fixed-u transition has zero second derivative. The squared norm of its first derivative is at most $85/256 + 23/(256d)$: the two original nonzero entries are bounded by $9/16$ and $1/8$, and the capacity-induced entries are bounded by $1/(16d)$. Combining this with the diagonal stage-cost bound gives D_t. Finally, positive interpolation along one coordinate has error between zero and $D_t h^2/8$. Applying the coordinate interpolations successively and adding the bounds proves (A.119). *Q.E.D.*

1.5. *Proof of the result A.61*

PROOF: Joint convexity, including the feasible graph, gives

$$Q_t^*(x, \bar{\pi}_t(x)) \le \sum_v w_v(x) Q_t^*(v, a_t(v)) \le I_n q_t^{\mathrm{sel},+}(x).$$

The lower interpolation bound gives $V_t^*(x) \ge I_n l_t(x) - \kappa_t$. Subtraction proves the uniform optimal-advantage bound e_t. The partial action derivatives of Q_t^* are bounded by Λ_t, so feasible implementation error adds at most $\Lambda_t b_t$. Adding the future policy loss and proceeding backwards proves the recursion. No convexity of $J^{\widehat{\pi}}$ is used. *Q.E.D.*

1.6. *Proof of the result A.50*

PROOF: Apply concavity to $f(a) - Ha^2/2$ and substitute the two lower endpoints. The displayed quadratic has derivative $D - K + 2Ks$. Minimizing it on the unit interval gives the three cases. Its coefficient of K is $-s(1-s)$, which proves the last assertion. *Q.E.D.*

1.7. *Proof of the result A.51*

PROOF: At $r = 1$, analytic endpoint enclosures and the minorant lemma give the assertion. Suppose it holds for $r - 1$. Each child interval encloses the true optimal continuation. If the numerical next state is an interval, expanding the queried center value by G_{r-1} times the maximum ℓ^1 displacement encloses all its possible values. Conditional-mean lower bounds and the semiconcavity remainder then give valid endpoint brackets for the true Q_r. The minorants cover the entire action interval. A downward capacity endpoint can omit only an interval of length equal to the certified capacity-rounding difference; subtracting Λ_r times that difference accounts for its possible improvement. The minimum upper endpoint is the cost of a feasible first action followed by the optimal future. This proves (i) by induction. Retaining earlier valid lower bounds preserves every conclusion.

For termination, an interval of length h has minorant at least its smaller lower endpoint minus $H_r h^2/8$. Its smaller upper endpoint is an available global upper bound. With the endpoint-width guard, an interval of length at most $h_\delta = \sqrt{3\delta/H_r}$ cannot

be responsible for a gap exceeding δ. Every refined parent is therefore longer than h_δ. Its children are longer than $h_\delta/5$. Apart from the at most two initial fragments made by a prediction, disjoint terminal fragments consequently have total number at most $5c(x)/h_\delta$, up to endpoint counting. Equation (A.101) is a conservative point budget. Bounded action-quantum rounding of a proposed quarter-to-three-quarter split preserves the one-fifth property whenever the quantum is smaller than $h_\delta/20$. Otherwise precision must be increased; a fixed-precision run is not allowed to ignore this requirement.

For (iii), let $\bar{x}$ be the acquired state. Its downward displacement is at most $d2^{-b}$ in ℓ^1, and $c(\bar{x}) \le c(x)$ preserves feasibility. At a fixed action the stage cost is $8/d$-Lipschitz and the transition has column sums at most $11/16$. Thus replacing $\bar{x}$ by x changes Q_r by at most $(8 + 11\beta d G_{r-1}/16)2^{-b}$. The corresponding value change is at most $dG_r 2^{-b}$. Add these two allowances to (i). Substituting the policy's own future yields the usual one-step loss recursion; backward induction gives (A.102). No policy-value oracle and no population claim about fitting are used. *Q.E.D.*

1.8. *Proof of the result A.41*

PROOF: A routed action is merely another feasible partition point. The endpoint and minorant induction in Theorem A.51 therefore applies without change. After each evaluation retain the smallest upper endpoint and the maximum of the old and newly valid global lower endpoints. Their difference cannot increase. The stopping test uses this difference, not a fitted value or a prediction error.

Put $h_\delta = \sqrt{3\delta/H_r}$. A partition interval of length at most h_δ has minimum minorant at least its smaller lower endpoint minus $3\delta/8$. The associated upper endpoint, with the stipulated width and capacity allowance, is within δ of that minorant. Such an interval cannot be responsible for an unresolved global gap. Every split parent therefore has length greater than h_δ. The one-fifth condition makes each of its children longer than $h_\delta/5$. Starting with one interval, disjoint terminal fragments consequently number at most $1 + 5c(x)/h_\delta$. Counting endpoints and allowing for rounding of this numerical bound gives the displayed conservative budget. Floating

failure is handled by the separately charged exact-rational procedure, not by this real-arithmetic count.

The first assertion about predictor evaluation follows from the order of the stopping test and the routing rule. For the last assertion, use backward induction on the remaining horizon, with the common analytic terminal kernel as base. At each nonterminal query, declined predictions leave the same conventional split, lower bound, witness and tie rule. Induction over partition refinements gives the same next query and returned endpoints; the induction over dates then gives the same implemented policy. Model-evaluation work can still differ, so transcript identity is not a zero-cost fitting claim. *Q.E.D.*

1.9. *Proof of Proposition A.42*

PROOF: Every retained lower endpoint is at least its true endpoint value minus w, and hence at least $f(a^*) - w$. On an interval of length at most h, the parabolic minorant is therefore at least $f(a^*) - w - Hh^2/8$. Subtracting the omitted-capacity allowance gives $L \geq f(a^*) - w - Hh^2/8 - \Lambda\Delta$; retaining an older, larger valid lower bound only improves this inequality. The upper endpoint at $\widehat{a}$ is at most $f(\widehat{a}) + w$. Semiconcavity and the one-sided first-order condition give $f(\widehat{a}) - f(a^*) \leq \lambda e + He^2/2$. Subtract the lower bound to prove (A.87). At an interior optimum the derivative is zero. The example $f(a) = a$ on $[0, c]$ shows that the boundary linear term is needed. Finally, retaining bounds makes $U' \leq U$ and $L' \geq L$; expansion proves (A.88). *Q.E.D.*

1.10. *Proof of the result A.38*

PROOF: For each drawn bin sequence, conditional integration over its continuous realizations places the exact path expectation between the interval endpoints. Uniform mixing over the equal bins recovers the original continuous law. The lower endpoint expectation is thus no larger than D_i, and the upper endpoint expectation is no smaller. Apply the bounded empirical-Bernstein inequality of Maurer and Pontil (2009) to the endpoint samples and take the union bound, with outward arithmetic included in the mean, variance and radius. Conditional validity for every

$\mathcal{H}$ implies unconditional validity by integration. On this one simultaneous event, all ensuing deterministic comparisons of the covered quantities hold together. *Q.E.D.*

1.11. *Proof of the result A.39*

PROOF: Subtract the gain intervals from k and compare the resulting interval endpoints with zero. Taking the image of $[l, u]$ under the absolute-value map gives (A.83). The assertions hold simultaneously for every nonnegative fee on the original coverage event. *Q.E.D.*

1.12. *Proof of the result A.40*

PROOF: Subtract the common J_{b_0}. The selected implementation has net difference at most $U(\lambda)$, and the best catalogue implementation has net difference at least $L(\lambda)$. This proves the regret bound pointwise on the one simultaneous event and hence for every λ. A minimum of finitely many affine functions changes its active affine expression only at an intersection. For $\lambda_2 > \lambda_1$, sum the two minimizing inequalities to obtain $(\lambda_2 - \lambda_1)(s_{\widehat{b}(\lambda_2)} - s_{\widehat{b}(\lambda_1)}) \le 0$. Since s_b increases strictly with b, the precision comparative static follows. *Q.E.D.*

2. PROOFS FOR REFERENCE-CENTERED RISK QUALIFICATION

2.1. *Centering and the original Bellman optimum*

For any bounded z, substituting $f_{t+1} + z$ in (10) and subtracting the finite number $\rho(f_{t+1})$ proves (12). Each dual objective at f_{t+1} is at most its supremum, giving $k^f \ge 0$. Approximating that supremum gives $\inf_q k^f(q) = 0$ without an attainment assumption. Since each q is a probability, adding a constant b to z adds b to $\mathcal{E}^f(z)$. If $z \le w$, every dual objective at z is no larger than its counterpart at w, proving monotonicity. The inequalities $w - \|z - w\|_\infty \le z \le w + \|z - w\|_\infty$ prove nonexpansiveness.

PROOF OF THEOREM 4: Let $d_t^\pi = V_t^\pi - f_t$ and $d_t^* = V_t^* - f_t$. The original recursions and (12) give the exact identities

$$d_t^\pi(x) = r_t(x, \pi_t(x)) + \beta_t \mathcal{E}^f_{t,x,\pi_t(x)}(d_{t+1}^\pi),$$

$$d_t^*(x) = \inf_{a \in A_t(x)} \{r_t(x,a) + \beta_t \mathcal{E}_{t,x,a}^f(d_{t+1}^*)\}. \tag{1}$$

Both have terminal value $g - f_T$. Suppose $d_{t+1}^\pi \le U_{t+1}$. Monotonicity, nonnegative β_t, the upper residual inequality and (13) imply $d_t^\pi \le U_t$. Similarly $L_{t+1} \le d_{t+1}^*$ implies $L_t \le d_t^*$. Backward induction establishes both statements. Feasibility of the supplied complete policy implies $V_t^* \le V_t^\pi$, giving (14). The argument takes place at each fixed parameter vector; applying it to every vector satisfying the premises proves the class statement without assembling incompatible continuations.

Exact equality recursions coincide with (1), proving the exactness assertion by induction. For the scalar comparison, set $a_t = \ell_t + b_t^-$ and $b_t = u_t + \gamma_t + b_t^{\pi,+}$ from the signed assumptions. Every original action has residual at least a_t, since its numerical action value is at least the numerical Bellman infimum. The implemented action has residual at most b_t. Define $\underline{h}_T = d^-$, $\overline{h}_T = d^+$, and

$$\underline{h}_t = a_t + \beta_t \underline{h}_{t+1}, \qquad \overline{h}_t = b_t + \beta_t \overline{h}_{t+1}.$$

Since $\mathcal{E}^f(b) = b$ for constants, these are feasible lower and upper certificates. Exact backward evaluation is order preserving, so $L_t \ge \underline{h}_t$ and $U_t \le \overline{h}_t$ for exact residuals and terminal discrepancies. Their date-zero difference is at most (A.7).

For two policy supersolutions, their pointwise minimum remains a supersolution: the operator at the minimum of their successor bounds is at most its value at either bound. The corresponding pointwise maximum of two optimum subsolutions remains a subsolution by the reverse argument. Finite combinations therefore preserve an already valid scalar guarantee. No claim about improved bounds from an uncontrolled numerical optimizer is needed. *Q.E.D.*

2.2. *Risk occupations and certificate orientation*

PROOF OF PROPOSITION A.13: For fixed π, (1) is the Bellman equation for a finite-horizon maximizer selecting a risk index j after observing the current state. Its reward is $r_{ts} - \beta_t k_{tsj}$, transition vector is q_{tsj}, and terminal reward is $g - f_T$. Finite maxima have measurable maximizers. Backward induction establishes a Markov maximizing choice at each date and state.

Any nonanticipating choice rule, including a history-dependent one, induces occupation masses satisfying (A.22). Conversely, for a feasible flow, choose j at (t,s) with probabilities y_{tsj}/m_{ts} when $m_{ts} > 0$; at zero-mass states choose an arbitrary feasible index. Induction on t recovers the declared masses. The corresponding expected sum of rewards is exactly the objective in (A.23). Thus the flow maximum equals the centered fixed-policy value. This representation is under the adversarial risk kernels and must not be called the physical occupation law.

Multiply each dual inequality by $B_t y_{tsj}$ and sum. The flow equalities cancel intermediate potential terms, leaving the primal objective at most $\mu^{\mathsf{T}} U_0$. The argument allows a zero discount: it uses $B_{t+1} = B_t \beta_t$ but never divides by either factor. Feasible primal and dual witnesses with equal objectives certify the maximum. For an optimum subsolution, a separate feasible risk witness for every original action supplies a lower bound on that action's supremum; a minimization over a restricted sampled action set does not supply the required all-action inequality. *Q.E.D.*

2.3. *Kernel exclusion and its sharp boundary*

For any $z \in \mathcal{D}$, the difference between the affine pieces indexed by q and q_0 is

$$(q - q_0)^{\mathsf{T}} z - k^f(q) + k^f(q_0).$$

A strict negative upper bound excludes q uniformly. This proves (A.24). Maximizing coordinate by coordinate gives the box formula. For a span ball, the coefficients sum to zero; adding a constant to z changes nothing. Normalize $\min_i z_i = 0$ and assign s to the positive coefficients and zero to the negative ones. The maximum is $s \sum_i (q_i - q_{0i})_+ = s\|q - q_0\|_1/2$.

As a sharp example, take two equally likely outcomes and average value at risk at level $1/2$. Then $\rho(v) = \max(v_1, v_2)$ and the dual vertices are e_1, e_2. At $f = (0,1)$ their centered penalties are $1, 0$. For $z = (2,0)$, the centered value is 1. Keeping only the maximizing kernel e_2 from the reference gives 0, which is invalid as an upper bound; dropping penalties gives 2, which is conservative but not exact. On the span ball of radius $1/2$, the first kernel is safely excluded. At radius 1, $z = (1,0)$ makes the two pieces equal, so the strict exclusion must retain the boundary. This example

proves both the necessity of allowing the risk kernel to change and sharpness of the strict threshold convention.

2.4. *Entropic duality without changing the objective*

Let p be a probability measure, f, z bounded, and $\lambda > 0$. Boundedness makes $Z_f = \int e^{\lambda f} dp$ strictly positive and finite, and makes p_f equivalent to p on its support. For $q \ll p$ with finite relative entropy,

$$\lambda^{-1}\mathrm{KL}(q\|p_f) = \lambda^{-1}\mathrm{KL}(q\|p) - \int f\, dq + \lambda^{-1}\log Z_f.$$

If q places mass outside the support of p, its penalty is infinite. For any bounded v, define $p_v = e^{\lambda v} p / Z_v$. Nonnegativity of $\mathrm{KL}(q\|p_v)$ gives

$$\int v\, dq - \lambda^{-1}\mathrm{KL}(q\|p) \le \lambda^{-1}\log Z_v,$$

with equality at $q = p_v$. This proves the variational representation, its centered penalty, and (A.25). It also shows exactly why using the fixed tilt p_f as an expectation is not generally exact: the centered evaluation remains entropic under p_f, rather than linear. For $\lambda \downarrow 0$, bounded convergence gives the expectation limit. Exponential and logarithmic evaluations in a numerical certificate require outward arithmetic; the rational average-value-at-risk experiments are not evidence of exact floating-point entropic computation.

2.5. *Occupation-sensitive precision and a strict improvement*

For fixed penalties and transitions, write the flow objective as $A(y)$ and the additional nonnegative residual contribution as $E(y) = \sum_{t,s} B_t e_{ts} m_{ts}$. If $H = \max_y A(y)$ and $H_e = \max_y \{A(y) + E(y)\}$, then

$$0 \le H_e - H \le \max_y E(y).$$

The upper inequality follows from $A(y) \le H$ for every feasible y; the lower follows from $E(y) \ge 0$. Since each occupation layer has total mass one, $\max_y E(y) \le \sum_t B_t \max_s e_{ts}$. This establishes (A.26). The argument assumes the penalties are held

fixed while residual uncertainty is enlarged. Altering the underlying stored continuation changes the penalties and requires requalification.

For a strict horizon example, let two states be absorbing under both actions, let action zero cost zero and action one cost one in both states, let $g = 0$, and let all discounts equal one. Every normalized monetary risk evaluation of these deterministic transitions agrees with evaluation at the absorbing state. Put $f_t(s) = (T - t)s$ for $s \in \{0, 1\}$ and choose action zero everywhere. Each date's implemented and minimum residuals are $(0, -1)$; the signed scalar account charges one at each date, yielding T. The exact state-dependent lower and upper recursions both equal $-(T - t)s$, giving zero policy loss at every state. There are two feasible actions and every action is checked. No state-dependent bias was cancelled without propagation: its entire state dependence was carried through the original transition at every date. This is consistent with, and sharpens rather than contradicts, the statewise-bias warning in Section B.

Finally, the finite backward recursion evaluates each stored transition support once for each retained risk piece at each date and state-action pair. Summing these support sizes yields the stated arithmetic count. A direct descendant recursion repeatedly evaluates the same subproblem; removing that repetition does not establish a polynomial bound in the number of continuous states, risk pieces, acquisition cells or rational bit length. The comparison is a representation-conditioned resource statement, and the full construction and support-qualification work remains separately charged.

3. CONTINUOUS INVESTMENT: PRIMITIVE PROOFS

3.1. *Curvature, the action constraint, and state moduli*

PROOF OF PROPOSITION 5: For $x \in X$, $0 \leq a \leq 1/4$ and $|z| \leq 1/32$, the transition remains in X. Indeed, the state part is increasing in both of its arguments, its maximum is $1/16 + 1/2 + 1/8$, and $b_j a \leq 1/8$. Its minimum after the shock is at least $1/32$, and its maximum is at most $27/32$. Thus every derivative bound below is used on its stated domain.

Both S and g are convex and continuously differentiable; their squared hinges have positive semidefinite second derivative on each smooth region. Direct differ-

entiation gives

$$|\partial_j g| \le \frac{10}{d}, \qquad |\partial_j S| \le \frac{15}{2d}, \qquad \nabla^2 S \succeq \frac{4}{d} I.$$

The first bound adds $5/d$ from the target term, $1/d$ from the cyclic difference term, and $4/d$ from the hinge. Only $-x_j x_{j+1}/16$ in F_j has nonzero second derivative. For a joint direction (v, w), the possibly negative part of the second derivative of $g(F)$ is

$$-\frac{1}{8}\sum_j \partial_j g(F) v_j v_{j+1} \;\ge\; -\frac{5}{4d}\|v\|_2^2,$$

since $\sum_j |v_j v_{j+1}| \le \|v\|_2^2$. The remaining term is the quadratic form of the positive semidefinite Hessian of g applied to the full state-action Jacobian. In particular, it includes the mixed state-action terms and is not discarded selectively. Since $h''(a) = 2 + 48a^2 \ge 2$, every fixed-shock objective satisfies (17). Integrating against a fixed feasible risk density preserves this bound. Taking the supremum over those densities preserves the same strong convexity. The bound extends across the hinge by continuity of the first derivative, or by integrating the almost-everywhere second-derivative inequality along a segment.

The graph $\{(x, a) : x \in X, 0 \le a \le A(x)\}$ is convex because A is affine. Partial minimization of the jointly convex objective over this graph proves convexity of V_1. Compactness and continuity supply an attained minimum. The positive capacity permits constraint qualification in the action variable.

For a fixed feasible density, the supporting strong-convexity inequality gives, for every (y, u),

$$Q_1(y, u) \ge W + p_x^\top (y - x_0) + p_a(u - a_0) + (u - a_0)^2.$$

For feasible u, adding $\lambda(u - A(y)) - \nu u \le 0$ to the right side preserves a lower bound. Minimize this augmented quadratic over all real u, an enlargement of the feasible interval. Completing the square gives (19). This proof uses neither equality in the risk supremum nor exact stationarity. Interval implementations bound the complete expression outwards, including its residual square and its slope.

For completeness we bound true value subgradients, rather than the potentially large slopes of arbitrary poor lower witnesses. A transition Jacobian column has

nonnegative entries summing to at most $11/16$. Therefore any state subgradient of a risk-maximizing one-date objective has coordinate magnitude at most

$$\frac{1}{d}\left(\frac{15}{2}+\beta\frac{55}{8}\right).$$

At an upper-bound action, the multiplier can be chosen with $0 \le \lambda \le \beta(10/d)\sum_j b_j = 15\beta/4$: the nonnegative derivative of h cannot increase the negative derivative which this multiplier offsets. At a lower-bound action take $\lambda = 0$. At an interior minimizer both action-bound multipliers vanish. Envelope subgradients thus have coordinates bounded by

$$\frac{1}{d}\left\{\frac{15}{2}+\beta\left(\frac{55}{8}+\frac{15}{32}\right)\right\} < \frac{15}{d}.$$

Where the risk function is nonsmooth, the same bound holds for convex combinations of active risk gradients and valid KKT multipliers. Integrating bounded subgradients along a segment, with the one-sided limit at the boundary of X, proves (18). *Q.E.D.*

3.2. *Exact one-date risk and outward construction*

Put $m = F(x,a,0)$, $R = 1/32$ and $D = 10d^{-1}\sum_j m_j s_j$. Since $\sum_j s_j = 0$, the hinge in $g(m+sz)$ is constant in z. The target and cyclic terms give exactly

$$g(m+sz) = g(m) + Dz + 5z^2. \tag{2}$$

Let $L = 2R\alpha$ and $h = \max\{-R, \min\{R-L, -D/10 - L/2\}\}$. The interval $[h, h+L]$ is a lowest-value interval of length L for the convex quadratic. The maximizing AVaR density is $(1-\alpha)^{-1}$ on its complement and zero on that interval. The two tail moments are

$$M_1 = -\frac{L(2h+L)}{2(2R-L)}, \qquad M_2 = \frac{2R^3 - L(3h^2+3hL+L^2)}{3(2R-L)}.$$

Hence the risk equals $g(m) + DM_1 + 5M_2$. This also follows from the threshold representation of AVaR, with the threshold chosen at the boundary of the retained

lower interval. Endpoint minima and equal boundary values are included by the clipping rule.

For a lower witness, any dyadic $\widehat{h} \in [-R, R-L]$ defines an exactly feasible excluded interval and a feasible tail density. Differentiating its objective holds $\widehat{h}$ fixed. This distinction is essential: a derivative of a changing approximate tail need not be the derivative of a supporting objective. Upper values use an outward enclosure of the maximizing h, while lower witnesses use the separately retained feasible $\widehat{h}$. The lower witness therefore remains valid when the two differ.

At a fixed initial state, $F(x,u,z)$ is affine in u and z. A convex continuation has a global affine minorant at every successor anchor. For a density constant on a bin, integration of this affine minorant equals its value at the midpoint. Its action dependence is affine. Adding the strong-convexity minorant of h gives (20). A fitted continuation is used only where its nonnegative convex readout makes this argument valid; no analogous assertion is made for an unrestricted nonconvex fit.

For a value upper bound, convexity places a fitted continuation below the larger endpoint value on a bin. For V_1, the midpoint upper value plus $15R/n$ bounds the whole bin by (18). If $l_i \le v(z) \le u_i$ on each bin, monotonicity of the original risk gives

$$\rho_\alpha\left(\sum_i l_i 1_{B_i}\right) \le \rho_\alpha(v) \le \rho_\alpha\left(\sum_i u_i 1_{B_i}\right).$$

The step-function risks are finite upper-tail means because the bin probabilities are exactly equal and $n(1-\alpha)$ is an integer. Their computation does not assume that v is constant on the bins.

3.3. *Complete policy and centered comparison*

PROOF OF THEOREM 6: Every fixed density is feasible for the original risk supremum. Every continuation support is below V_1 at every successor. Accordingly its all-action quadratic minimum is below V_0, irrespective of which feasible proposal supplied the anchor. The binwise upper calculation dominates the value of the entire terminal controller; monotonicity and cash invariance charge at most $\beta\varepsilon_1$ for

its terminal optimality allowance. It follows that $\underline{V}_0 \le V_0 \le J_0^{\widehat{\pi}} \le \overline{J}_0^{\widehat{\pi}}$. Subtracting proves (21).

For the centered statement take any lower and upper continuation functions ℓ, u furnished by the preceding supports and bounds. Their differences from f give admissible error functions. The exact centered identity cancels the common $\rho_\alpha(f)$ before any numerical enclosure. The preceding lower witness is a minorant of the resulting all-action expression; the preceding policy upper envelope is a majorant of the resulting policy expression. They are sufficient subsolution and supersolution inequalities in the general theorem. Computing them jointly is an implementation of that comparison, not a new objective. Loose global residual constants remain a valid alternative, but need not replace these sharper local witnesses. *Q.E.D.*

Whole-cell residual interpolation. On a state cell with vertices v_i, choose its multilinear convex weights $\omega_i(y)$. If f and V_1 are convex, their vertex upper enclosures interpolate to upper bounds. A fixed affine support L_V of V_1 satisfies

$$V_1(y) - f(y) \ge L_V(y) - \sum_i \omega_i(y)\overline{f}(v_i) = \sum_i \omega_i(y)\{L_V(v_i) - \overline{f}(v_i)\}.$$

Its right side is at least the smallest vertex difference. Maximize that lower bound over the stored supports. Reversing V_1 and f, using upper values of V_1 and lower supports of f, gives the upper residual bound. The finite maximum or minimum over anchors and cells is taken after these whole-cell inequalities. No interpolation equality for the unknown function is asserted.

Strict tail exclusion. Suppose $u_i < \tau$ and the union H of $k = n(1-\alpha)$ other bins has $v(z) \ge \tau$ throughout. It has probability at least $1-\alpha$. If a feasible maximizing density assigned positive mass to B_i, some capacity $q < (1-\alpha)^{-1}$ would remain on H. Moving equal mass from B_i to H strictly increases its risk objective, a contradiction. Equality yields only weak improvement and cannot support exclusion. Bin bounds used to construct H must come from the unpruned calculation. The argument covers a continuum of densities without enumerating all of them.

3.4. *Statistical unit and numerical allowance*

For two complete policies that share the terminal controller and differ only in their first action, the stage-action difference is at most $5/64$. Since $\sum_j b_j = 3d/8$ and the largest feasible action difference is $1/4$, (18) gives a future-value difference of at most $45/32$. Terminal-controller errors take values in $[0, \varepsilon_1]$, so their difference is at most ε_1, not twice that allowance. Monotonicity and cash invariance give the deterministic paired support

$$|J^A(x) - J^B(x)| \le B := 5/64 + \beta(45/32 + \varepsilon_1).$$

This support is established before observing a workload. It is not replaced by an observed sample range.

For n independently sampled initial states and paired cost intervals $[D_i^-, D_i^+]$, intersect each interval with $[-B, B]$. For any fixed center c computed from these intervals,

$$\frac{1}{n-1}\sum_i (D_i - \overline{D})^2 \le \frac{1}{n-1}\sum_i \max\{|D_i^- - c|, |D_i^+ - c|\}^2 =: \bar{s}^2.$$

Applying the two-sided empirical Bernstein inequality (Maurer and Pontil, 2009) to each of twelve fixed contrasts, followed by a union bound, is conservative with $\log(960) < 7$. A simultaneous sampling radius is

$$r_n = \sqrt{14\bar{s}^2/n} + \frac{98B}{3(n-1)}.$$

The resulting interval is $[n^{-1}\sum_i D_i^- - r_n, n^{-1}\sum_i D_i^+ + r_n]$, evaluated outwards. The calibration selector is fixed independently of these rows, and all pairwise contrasts are retained. Innovation bins, fitting labels, repeated sorting trials and repeated executions of an unchanged stream are not additional independent observations.

3.5. *Proof of the horizon-uniform construction*

PROOF OF PROPOSITION A.14: The primitive state cost has ℓ^1 Lipschitz modulus at most $8/d$. The transition's state Jacobian has nonnegative entries with every column sum at most $11/16$. Each control exposure column has ℓ^1 norm $3d/8$. On

the outer action rectangle $[0,1/4]^m$, each component of the action-cost gradient is at most $13/16$. The scalar and vector transitions remain in the cube on that rectangle: for the vector case their range is contained in $[1/64,59/64]^d$. These bounds therefore remain applicable during the state and action comparisons even when an intermediate action is not feasible at an intermediate capacity.

Capacity intervals and capacity simplexes have Hausdorff modulus $1/(8d)$ in the action ℓ^1 norm. In the vector case, when capacity decreases, scale an action radially to the smaller capacity; the required total action displacement is no larger than the capacity change. Monotonicity and cash invariance make AVaR nonexpansive in the supremum norm. It follows that any continuation with state modulus L gives a Bellman value modulus at most

$$\frac{8}{d}+\beta\frac{11}{16}L+\frac{1}{8d}\left(\frac{13}{16}+\beta\frac{3d}{8}L\right)=\frac{1037}{128d}+\beta\frac{47}{64}L.$$

Substitution of $L=27/d$ gives $\{1037/128+705\cdot 27/1024\}/d<27/d$. The terminal cost has modulus $10/d$. This proves the uniform value modulus by backward induction. The same calculation applies to an arbitrary Lipschitz fitted continuation; convexity of that fit is not required. Consequently every own-future cone may be constructed with the common primitive modulus $27/d$.

For convexity of the original values, assume inductively that V_{t+1} is convex and its coordinate subgradients have absolute value at most $27/d$. For a smooth such function, the negative contribution from the bilinear state transition is at worst $27\|v\|_2^2/(8d)$; the Jacobian–Hessian–Jacobian term is positive semidefinite in the full state-action direction. Adding stage curvature yields

$$\frac{1}{d}\left(4-\beta\frac{27}{8}\right)\|v\|_2^2+\frac{7}{4}\|w\|_2^2=\frac{107}{128d}\|v\|_2^2+\frac{7}{4}\|w\|_2^2.$$

The action bound includes the vector cost interaction, whose off-diagonal Hessian entries equal $1/4$. For a nonsmooth convex continuation, mollify on a neighborhood of the transition image, which is a fixed positive distance from the cube boundary. Mollification preserves convexity and coordinate Lipschitz bounds there. The preceding uniform curvature inequality passes to the limit. Integration under any fixed feasible innovation density and the supremum over those densities preserve

it. Partial minimization over the convex capacity graph proves convexity of V_t. This establishes the induction without assuming differentiability of the resulting risk-averse values.

The tensor state cover has ℓ^1 covering radius at most $d/(2n_x)$. Hence the two-sided cone term is at most $27/n_x$, including the terminal cone formed from exact terminal labels. The action objective modulus is bounded by

$$D_a \le \frac{13}{16} + \beta \frac{3d}{8} \frac{27}{d} = \frac{1319}{128}.$$

The scalar feasible net has radius no larger than $A(x)/(2m)$. The vector simplex lattice has ℓ^1 radius at most $A(x)/m \le 1/(4m)$. To verify the latter, scale by $m/A(x)$, take the coordinate floors, and, when they do not exhaust the integer capacity, compare the two points obtained by rounding one coordinate up. One of their ℓ^1 errors is at most one. If the floors exhaust capacity, the point is already a lattice point. Thus the common action-net contribution is at most $1319/(512m)$.

On one innovation rectangle, the maximal ℓ^1 displacement from its midpoint under the vector transition is $d(1/(32n_z) + 1/(64n_z))$. A $27/d$-Lipschitz continuation therefore varies by at most $81/(64n_z)$. Applying the original risk's nonexpansiveness and then discounting bounds the objective integration error by $e \le 1215/(1024n_z)$. The scalar case is covered by the same bound. Exact AVaR of the midpoint step function is computed by sorting the n_z^s equally weighted values; $4 \mid n_z$ ensures integral tail counts for both risk levels. The innovation law remains continuous: the bound on e is the explicit difference between its risk and the step-function risk.

The cone proof also applies to the present risk recursion. Indeed, for any bounded u, v and scalar c, monotonicity and cash invariance give $\|\rho(u) - \rho(v)\|_\infty \le \|u - v\|_\infty$ and $\rho(u + c) = \rho(u) + c$. Feasible-action minima preserve order and this bound. Thus label, action-net and continuation-cover errors propagate by the same discounted subsolution and supersolution induction; no linear expectation or frozen maximizing density is used in that induction. The graph and action moduli established above verify its primitive hypotheses for every own-future continuation. Theorem 2, with exact feasible repair and no additional label-arithmetic error, now

bounds each nonterminal residual sum by

$$b := 27/n_x + 1319/(512m) + 1215/(512n_z).$$

The terminal cone contributes at most $27/n_x$. Thus the finite-horizon bound is $\sum_{t<T} \beta^t b + \beta^T 27/n_x \le b/(1-\beta)$, which is (A.27). The displayed sufficient resolutions divide the approximation allowance into three equal discounted shares. With the same uniform moduli, the sensor, repair and action-rounding coefficients in (9) are bounded independently of the horizon; their discounted sum is at most sixteen times the one-date allowance. Finite sensor and action precision can therefore fill a separately reserved positive share of the target. *Q.E.D.*

Work, storage, and exact-number growth. Every date uses S targets, at most A_m feasible actions per target and n_z^s midpoint queries per action. A dense cone query scans S labels in $O(dS)$ rational operations, giving the stated dense count. The original-owner transform costs $O(dS)$ per date; cell location and the exact off-grid owner query give the compiled count. Sorting costs $O(TSA_m n_z^s \log(n_z^s))$ comparisons and is charged in addition, not absorbed into an unmeasured primitive oracle. Stored labels and witnesses occupy $O(TSd)$ rational words, with their bit lengths charged separately.

All fixed meshes, costs, transition coefficients, risk weights and cone moduli in this construction are rational. Their finitely many primitive denominators divide a fixed integer D whose logarithm is $O(\log(dn_x m n_z))$. At a new date, a stored label is combined with fixed-denominator primitive values, cone distances, tail averages and the rational discount. A common-denominator implementation therefore increases the denominator exponent by only a fixed amount per date. Maxima, minima, comparisons and exact feasible repairs do not increase it exponentially. Bounded discounted stage costs and the fixed cone modulus also bound label magnitudes uniformly in the horizon. The resulting numerator and denominator bit lengths are $O((T+1)\log D)$. Arithmetic on these integers is not assigned constant bit cost. The statement is a fixed-accuracy constructive horizon bound with an explicit state and innovation curse, not a measured arithmetic crossover or a full implementation benchmark.

4. DECISION-SCALE QUALIFICATION: PROOFS

PROOF OF PROPOSITION A.5: For every real threshold η, the function $\phi_\eta = (v - \eta)_+$ is convex and K-Lipschitz. Conditional Jensen and the chord inequality give $M_n\phi_\eta \le P\phi_\eta \le E_n\phi_\eta$. Apply the AVaR threshold formula $\inf_\eta\{\eta + (1-\alpha)^{-1}\mu\phi_\eta\}$ for each of the three probability measures. Taking infima preserves the two orders and proves the enclosure. In particular, no interchange of an action infimum and a risk supremum occurs.

Here is a uniform bound which remains valid at kinks. Let $h = 2R/n$. Choose one-sided slopes of ϕ_η at each bin's outer endpoints. The slope increases across each bin; its total increase over the whole interval is at most $2K$. The difference between the endpoint average and midpoint value on bin i is no greater than $h/4$ times the slope increase across that bin. This follows by integrating the left and right slopes over the two half-bins. Choose the left endpoint right derivative and right endpoint left derivative; jumps at shared endpoints may be omitted, which only decreases the sum of within-bin slope increases. Therefore

$$0 \le E_n\phi_\eta - M_n\phi_\eta \le \frac{h}{4n}(2K) = \frac{KR}{n^2}.$$

For completeness, the half-bin expression equals one half the integral of the difference between the right and left slopes at equal distances from the midpoint, so its upper bound also follows directly without assuming a classical second derivative. The preceding uniform bound passes through the two infima and gives (A.11). Monotonicity and cash invariance add $e_- + e_+$ for endpoint error. They also charge the actual terminal controller's one-sided error only on the upper side.

For a product law, apply the one-coordinate midpoint–endpoint comparison successively, holding the other coordinates fixed. Products of positive averaging operators preserve inequalities and telescope. Each partially averaged positive-part function remains convex and has the same Lipschitz modulus in each unintegrated coordinate. Summing the one-coordinate error bounds and then taking the threshold infimum proves the product statement. This argument uses neither a discrete economic transition assumption nor differentiability of v. *Q.E.D.*

PROOF OF THEOREM A.7: Every service's regret belongs to $[0, \tau]$ at every possible initial state. The difference of two complete-policy values therefore belongs to $[-\tau, \tau]$, because the common original optimal value cancels. Each interval intersection retains the true paired difference. A difference of two intervals of width at most w has width at most $2w$. Intersecting with a valid support cannot increase that width.

Hoeffding's inequality for a variable of range length 2τ gives

$$\Pr\{|\overline{D} - \mathbb{E}D| > r\} \leq 2\exp\{-Nr^2/(2\tau^2)\}.$$

A union bound across the H fixed contrasts gives the stated coverage. Pairwise contrasts need not be mutually independent; only the initial states within each contrast must be independent, conditional on the frozen preparation. Replacing the unobserved sample mean by its numerical lower and upper bounds yields (A.12). Their difference is at most $2(r + w)$. This is an application of the established concentration inequality (Hoeffding, 1963), not a claim to a new concentration principle.

For the chosen rational parameters, $Nr^2/(2\tau^2) = 8$. The inequality $24e^{-8} < 1/20$ can be checked entirely with rational arithmetic: the partial sum $\sum_{j=0}^{11} 8^j/j!$ is greater than 480. Equation (A.13) is an exact rational comparison. In numerical classification the margin is rounded inward and the reported confidence endpoints outward. No observed range or variance is used to choose the radius. *Q.E.D.*

Totalization of the admitted service. The first-date objective with the original V_1 has action Lipschitz constant less than 7: the primitive action derivative is at most $3/4$, and the discounted continuation contribution is at most $(15/16)(15/d)(3d/8)$. The feasible interval has length at most $1/4$. A uniform net with $m = 1024$ intervals therefore gives an all-action lower value by subtracting $7/(8m)$ from the smallest grid cost lower bound. The grid point with the smallest upper value supplies a feasible policy upper value.

The exact fallback uses $n = 256$ innovation bins and the preceding convex-order rule. Exact one-date rational supports have gaps at most 2^{-30}; the terminal controller contributes 2^{-20}. The resulting quadrature width, including these allowances, is

below $1/8192$ for both risk levels. Adding $7/(8m)$ leaves the original-optimum bound below $1/256$. Downward rounding of the first action to 40 bits costs at most 72^{-40} on each side and fits strictly inside both remaining allowances. All values, risk weights, feasible actions and endpoint computations in this final construction are rational, and the terminal strong-convexity bisection reaches its stated gap in finitely many steps. Thus the fallback defines a finite valid action at every dyadic initial observation in the target distribution. For arbitrary real initial states, the separate acquisition transfer remains required; the confirmation theorem here concerns the declared dyadic initial law.

The floating implementation may reach the two targets much sooner. That does not permit removing the exact fallback from the policy definition, treating rejected proposals as missing observations, or asserting that the fallback was exercised when the actual counter is zero. Hardware or resource failure aborts the experiment and is recorded as a failure, not a change in its conditional sampling law.

PROOF OF PROPOSITION A.8: Subtract the two complete acquisition-and-deployment accounts. Using the upper policy, acquisition and fee bounds and the lower computational saving bounds above (A.14) by $m(d^+ - \lambda s^-) + \lambda k^+ + F^+$. Nonpositivity gives (A.15). Solving this scalar inequality proves the integer threshold and its sign qualifications. The separate quality inequality depends on policy cost rather than price and cannot be deduced by increasing λ. A common audit cost cancels only when it is actually common and purchased in both alternatives; otherwise its incremental charge remains in k^+ or F^+. *Q.E.D.*

PROOF OF COROLLARY A.6: Write $h = 2R/n$ and $k = n(1-\alpha)$. The two roots of $v_n - \eta_n$, where $\eta_n = K\alpha R$, are $c_n - \alpha R$ and $c_n + \alpha R$. They lie strictly inside the domain and at two distinct bin midpoints, since αn is an even integer. Under M_n the mass strictly above η_n is $(k-1)/n$ and the mass at η_n is $2/n$. Under E_n there is no mass at η_n, and the mass above it is exactly k/n. Hence η_n minimizes the AVaR threshold objective under both measures, by its left and right slope conditions.

The hinge $\phi = (v_n - \eta_n)_+$ is affine on every bin except the two whose midpoints are its roots. On either exceptional bin, the endpoint average minus the midpoint value is $Kh/4$. Each bin has weight $1/n$. The two threshold minima therefore differ

by

$$(1-\alpha)^{-1}(E_n\phi - M_n\phi) = \frac{1}{1-\alpha}\frac{2}{n}\frac{Kh}{4} = \frac{KR}{(1-\alpha)n^2}.$$

All equalities use the endpoint mixture's half weights at the outer boundaries. This explicit sequence establishes sharpness of the stated enclosure on its function class; it does not assert that the investment value itself equals v_n. *Q.E.D.*

5. ACTION-ALIGNED CERTIFICATES AND SELECTION

5.1. *Proof of Theorem 7*

Taking the infimum of the second inequality in (23) and subtracting it from the first gives the gap bound. Substitution of $b = a$ in that inequality shows $L \le Q(a) \le U$, hence $w \ge 0$. The minimum in (24) is at most zero because a is feasible, so $\Gamma \ge 0$.

For $z = b - a$ and the supplied $n \in N_A(a)$,

$$s^\mathsf{T} z = (s - \widehat{s} + \widehat{s} + n)^\mathsf{T} z - n^\mathsf{T} z \ge -(d+r)\|z\|_2.$$

The inequality uses $n^\mathsf{T} z \le 0$ and the triangle and Cauchy–Schwarz inequalities. Completing the square yields

$$-(d+r)\|z\|_2 + \frac{\mu}{2}\|z\|_2^2 \ge -\frac{(d+r)^2}{2\mu}.$$

Taking the minimum over the original feasible set proves (25). In one dimension, completion of the square centers the minimization at $a - s/\mu$, whose projection onto the interval gives the stated minimizer. This argument includes $a = 0$ and $a = A_0$ and does not differentiate a discontinuous actor. A lower numerical bound on the minimum only enlarges Γ, preserving validity. *Q.E.D.*

5.2. *Proof of the risk-witness lift*

Strong convexity of the stage cost gives

$$c(b) \ge c(a) + \nabla c(a)^\mathsf{T}(b-a) + \frac{\mu}{2}\|b-a\|_2^2.$$

For the common feasible risk witness, the dual representation and the continuation support imply

$$\rho(V \circ F(b)) \geq \int V(F(b,\omega))\, q(d\omega) - \psi(q) \geq \int \ell\, dq - \psi(q) + \left(\int G^{\mathsf{T}} v\, dq \right)^{\mathsf{T}} (b - a).$$

Adding the inequalities gives (26). The same q must be feasible for every b in the comparison. If an upper penalty endpoint replaces $\psi(q)$, the intercept decreases and remains valid. Errors in a vector integral must be retained as a set of possible slopes or transferred uniformly to the intercept using the bounded action displacement. Choosing a favorable slope from an interval without this transfer is invalid. The complete policy upper U must separately dominate the full risk envelope, not just the selected lower witness. *Q.E.D.*

5.3. *Dated action losses and own-future continuations*

Suppose first that a feasible complete policy has, at every date and state,

$$Q_t(V_{t+1}^*; x, \pi_t(x)) - V_t^*(x) \leq \gamma_t$$

with nonnegative constants γ_t. If $V_{t+1}^{\pi} \leq V_{t+1}^* + e_{t+1}$, monotonicity and cash invariance give

$$V_t^{\pi} \leq Q_t(V_{t+1}^*; x, \pi_t(x)) + \beta_t e_{t+1} \leq V_t^* + \gamma_t + \beta_t e_{t+1}.$$

Backward induction with $e_T = 0$ proves $V_0^{\pi} - V_0^* \leq \sum_{t<T} B_t \gamma_t$. A certified terminal-controller allowance is inserted at its actual date. No product of tail multipliers is used to propagate a constant.

This statement does not assume that an NBO receives V_{t+1}^* as a free training target. A support built only for its own attained f_{t+1} bounds $Q_t(f_{t+1})$, not $Q_t(V_{t+1}^*)$. It must therefore enter the one-sided or centered residual sandwich with a certified continuation discrepancy. The investment construction instead supplies genuine original-V_1 supports and a separately bounded terminal controller. These are two legitimate implementations of the same original-optimum obligation; a small own-future greedy gap alone is not that obligation.

5.4. *Proof of Theorem 8*

Conditional on acquisition, the menu and all its functions are fixed. For any one h and any loss $Z_h \in [0,1]$, Hoeffding's inequality gives

$$\Pr\left(\left|n^{-1}\sum_i Z_h(X_i) - \mathbb{E}Z_h(X)\right| > e \mid \mathcal{A}\right) \le 2e^{-2ne^2}.$$

A union bound over the M candidates, at $e = e_n$, gives a common event of conditional probability at least $1-\delta$. On that event, for every h,

$$\mathbb{E}Z_{\widehat{h}} \le \widehat{Z}_{\widehat{h}} + e_n \le \widehat{Z}_h + e_n \le \mathbb{E}Z_h + 2e_n.$$

Use $Z_h = R_h$ to obtain (28). Use $Z_h = \ell_h$ and the pointwise inequality $R_h \le \ell_h$ to obtain the surrogate assertion. These are separate choices of loss; claiming both events simultaneously requires a corresponding union allocation. The work implication follows by expectation of its verified pointwise cap. For direct work minimization use $Z_h = W_h / W_{\max}$ and multiply the bound by $W_{\max}$. Bounds exceeding the trivial support limit may be intersected with it. Validation and acquisition costs do not disappear from total workload cost because they are absent from the per-context random variable. *Q.E.D.*

A minimal counterexample explains why a global residual objective cannot replace this loss. Let the action set be $[-1,1]$, the true objective $Q(a) = a^2$, and consider two continuation perturbations added to this objective. The first is $e_0(a) = a^2$, with oscillation one and unchanged minimizer zero. The second is $e_1(a) = \eta a$, with oscillation $2\eta < 1$ for $0 < \eta < 1/2$, but its minimizer is $-\eta/2$ and its true regret is $\eta^2/4 > 0$. Thus even strict oscillation reduction can worsen the delivered action. This example is an explanatory construction, not a diagnosis inferred for each retained investment cell.

6. NONUNIFORM AND RANDOMIZED AVAR INFORMATION

6.1. *Proof of the nonuniform lower bound*

Let q, L, t be as in Theorem 9 and answer every query with

$$f_0(z) = \frac{K}{2L}(z-q)_+^2.$$

The function is convex, nondecreasing and K-Lipschitz. Collect the queried points strictly inside (q, R), insert the two endpoints, and write the resulting nonempty interval lengths as $h_1, \dots, h_s$, with $s \le N + 1$. Let $f_1 = f_0$ to the left of q and equal its chord on each of those intervals. Successive chord slopes are nondecreasing and belong to $[0, K]$, including the junction at q. Therefore $f_1 \in \mathcal{C}_{K,R}$ and is nondecreasing. The functions agree at every query, and induction on the adaptive transcript proves that a deterministic procedure has the same output on both.

Because the functions are continuous and nondecreasing and P has positive density, their upper-tail risks are $t^{-1} \int_q^R f(z)p(z)\,dz$. Flat portions cause no difficulty: the quantile formula integrates the appropriate fraction of their mass. On an interval of length h, the unweighted integral of the chord error is $Kh^3/(12L)$. Since $p \ge m$,

$$\mathrm{AVaR}_\alpha^P(f_1) - \mathrm{AVaR}_\alpha^P(f_0) \ge \frac{mK}{12Lt} \sum_{i=1}^{s} h_i^3 \ge \frac{mKL^2}{12t(N+1)^2}.$$

The last inequality follows from $\sum_i h_i = L$ and convexity of the cube. A sound common interval contains both values. Finally $t = \int_q^R p \le M_p L$, so $L \ge t/M_p$, giving the second lower bound. The worst-case error of a point-valued deterministic procedure is at least half the first width. *Q.E.D.*

6.2. *Proof of the nonuniform upper bound*

Write the uniform cell length as $h = 2R/n$. For a cell $[a, b]$, let its mass be p_i and its conditional mean be μ_i. For any convex function g,

$$p_i g(\mu_i) \le \int_a^b g(z)p(z)\,dz \le p_i \left\{ \frac{b - \mu_i}{h} g(a) + \frac{\mu_i - a}{h} g(b) \right\}.$$

For $g_\eta = (f - \eta)_+$, these are valid for every threshold η; g_η is convex and K-Lipschitz. If $a < \mu_i < b$, the difference of the right and left expressions is

$$p_i \frac{(\mu_i - a)(b - \mu_i)}{h} \left\{ \frac{g_\eta(b) - g_\eta(\mu_i)}{b - \mu_i} - \frac{g_\eta(\mu_i) - g_\eta(a)}{\mu_i - a} \right\}.$$

The factor before the braces is at most $p_i h/4 \le M_p h^2/4$. The slope difference is bounded by the variation of the one-sided derivative within the cell. Summing these nonnegative variations over the disjoint cell interiors is at most the total

derivative variation, which is at most $2K$. Boundary means follow by continuity (and do not occur with strictly positive density). Hence the summed upper-minus-lower expectation is at most $M_p K h^2/2 = 2M_p K R^2/n^2$, uniformly in η.

Add η, divide each expectation by t, and minimize over η in the AVaR variational representation. The numerical lower objective lies below the original one and the numerical upper objective above it. Their difference is uniformly bounded by (30); taking infima preserves both enclosure and this width bound. The numerical rules require the n mean values and the $n+1$ endpoint values of f, with repetitions shared. Known moments specify their nonnegative weights, summing to one. An approximate moment rule requires a separately proved perturbation allowance and is not covered by an uncharged exact-law oracle. *Q.E.D.*

6.3. *Proof for pathwise-sound random intervals*

Use the same fixed f_0 as above, before drawing a random tape. Outside the single null set in the soundness definition, fixing the tape gives a deterministic adaptive procedure that is sound for all inputs. The preceding chord construction therefore forces the stated width on its transcript from f_0. The chord competitor may vary with the tape; f_0 does not. Taking expectation gives the same lower bound on expected width. A procedure whose bad-tape set depends on the input need not be sound for that tape's chord competitor, which is why the argument proves no such assertion for ordinary $1-\delta$ coverage. *Q.E.D.*

6.4. *Proof of the randomized point-estimation lower bound*

For the uniform law set $q = -R + 2R\alpha$, $L = 2R(1-\alpha)$ and $s = z - q$. Partition the tail into $2N$ equal intervals of length $h = L/(2N)$. Let $\psi(u) = u^2(1-u)^2$ on $[0,1]$, and extend it by zero outside this interval. For independent signs $\theta_j \in \{-1,1\}$, define on the tail

$$f_\theta(q+s) = \frac{Ks^2}{4L} + \frac{Kh^2}{8L}\sum_{j=1}^{2N} \theta_j \psi\left(\frac{s-(j-1)h}{h}\right),$$

and put $f_\theta = 0$ to the left of q. We verify that this entire finite family belongs to the announced class. Both ψ and ψ' vanish at zero and one, so the functions and

first derivatives match at every cell boundary and at q. On each cell, $\psi''(u) = 2 - 12u + 12u^2 \in [-1, 2]$. Consequently

$$f''_\theta \geq \frac{K}{2L} - \frac{K}{8L}\,2 = \frac{K}{4L} > 0.$$

The derivative starts at zero and ends at $K/2$, since bump derivatives vanish at both tail endpoints. It is therefore everywhere in $[0, K/2]$. The full function is convex, nondecreasing and K-Lipschitz; second derivatives at junctions are not needed.

Since $\int_0^1 \psi(u)\,du = 1/30$, the original AVaR of this monotone function equals a known baseline plus

$$b_N \sum_{j=1}^{2N} \theta_j, \qquad b_N = \frac{Kh^3}{240L^2} = \frac{KL}{1920N^3}.$$

A point query inside one open cell depends only on that cell's sign; a boundary query or query outside the tail reveals none. Conditional on the algorithm's random tape and its entire adaptive transcript, the signs in unqueried cell interiors remain independent and equally likely. This follows inductively: the next location is measurable with respect to already revealed information and the tape, not unqueried signs. At most N of the $2N$ signs have been queried. The conditional variance of the target risk is therefore at least Nb_N^2.

For any estimator measurable with respect to the transcript and independent algorithmic randomization, conditional mean-square error is no smaller than that conditional variance. Averaging over signs and randomization gives Bayes risk at least Nb_N^2. The largest risk over this fixed finite family is at least its average; hence the worst-case root mean-square error is at least

$$\sqrt{N} b_N = \frac{KL}{1920N^{5/2}} = \frac{KR(1-\alpha)}{960N^{5/2}}.$$

This argument uses a fixed distribution over inputs rather than a tape-dependent adversarial input, and therefore applies to ordinary randomized point estimation. It supplies a lower bound only; an optimal upper algorithm for the whole convex class is not asserted. *Q.E.D.*

6.5. *Proof of the dated allocation identity*

For positive a_j, c_j, put $H = \sum_j a_j^{1/3} c_j^{2/3}$. Hölder's inequality gives

$$H = \sum_j (a_j/n_j^2)^{1/3} (c_j n_j)^{2/3} \le \left(\sum_j a_j/n_j^2 \right)^{1/3} \left(\sum_j c_j n_j \right)^{2/3}.$$

Thus feasibility implies $\sum_j c_j n_j \ge H^{3/2}/\sqrt{\varepsilon}$. Equality is attained by $n_j = (a_j/c_j)^{1/3}\sqrt{H/\varepsilon}$, for which the tolerance constraint binds. Upward integer rounding decreases each allowance and adds less than c_j per date. This argument assumes the stated linear per-resolution work law on a fixed graph; it does not substitute for pricing descendants or multidimensional tensor queries. *Q.E.D.*

7. DECISION SEPARATION AND EXACT ACQUISITION

7.1. *Proof of Theorem 11*

At fixed x, only prices between $h(x)$ and $v(x)$ induce a different action from the original optimum. At such a price, the regret is $|p - v(x)|$. Both thresholds lie in $[0,3]$, so integrating the triangle under the uniform price density gives

$$\frac{1}{3} \int_{\min(h,v)}^{\max(h,v)} |p - v| \, dp = \frac{(h-v)^2}{6}.$$

Integration over x proves the unconditional identity, with no approximation or simulated policy observation.

Write $f = v - 1 = (x_1 + x_2 - 1)_+$. Under independent uniform coordinates,

$$\mathbb{E} f = \frac{1}{6}, \quad \mathbb{E}[f \mid x_1] = \frac{x_1^2}{2}, \quad \mathbb{E}[f \mid x_2] = \frac{x_2^2}{2}, \quad \mathbb{E} f^2 = \frac{1}{12}.$$

The additive projection of f is $g = (x_1^2 + x_2^2)/2 - 1/6$. Its residual has conditional mean zero given either coordinate, hence is orthogonal to every square-integrable additive function. This proves optimality over the whole additive class, not merely over additive quadratics. The constant and centered coordinate terms are mutually

orthogonal, so

$$\|g\|_2^2 = \frac{1}{36} + 2\left(\frac{1}{20} - \frac{1}{36}\right) = \frac{13}{180}, \qquad \|f-g\|_2^2 = \frac{1}{12} - \frac{13}{180} = \frac{1}{90}.$$

The shifted optimum $1+g$ has range $[5/6, 11/6]$, satisfying the price-support restriction. Equation (34) gives $1/540$.

For the quadratic projection use the orthogonal shifted Legendre polynomials $P_0 = 1$, $P_1(u) = 2u-1$, $P_2(u) = 6u^2 - 6u + 1$, with squared norms $1/(2j+1)$. Direct integration on the upper triangle gives

$$\Pi_2 v = \frac{7}{6} + \frac{1}{4}(P_1(x_1) + P_1(x_2)) + \frac{1}{12}(P_2(x_1) + P_2(x_2)) + \frac{3}{10}P_1(x_1)P_1(x_2).$$

To verify these coefficients without assuming the answer, the monomial normal equations use basis $(1, x_1, x_2, x_1^2, x_1x_2, x_2^2)$, Gram entries $G_{(i,j),(k,l)} = 1/\{(i+k+1)(j+l+1)\}$, and right side

$$b_{ij} = \frac{1}{(i+1)(j+1)} + \int_0^1 \int_{1-x_1}^1 x_1^i x_2^j (x_1 + x_2 - 1)\, dx_2\, dx_1.$$

The Gram matrix is positive definite because a nonzero polynomial cannot vanish almost everywhere on the square. Substituting the coefficients in (37) gives $G\theta = b$ exactly; the complete rational matrix, vector and zero residual are supplied in the executable evidence. Orthogonality alternatively gives

$$\|\Pi_2 f\|_2^2 = \frac{1}{36} + 2\frac{(1/4)^2}{3} + 2\frac{(1/12)^2}{5} + \frac{(3/10)^2}{9} = \frac{37}{450}, \quad \|f - \Pi_2 f\|_2^2 = \frac{1}{900}.$$

The Hessian of (37) has eigenvalues $11/5$ and $-1/5$. Its interior critical point $(3/11, 3/11)$ is a saddle. Checking each edge gives the minimum $143/150$ at $(3/5, 0)$ or $(0, 3/5)$ and the maximum $32/15$ at $(1,1)$. Thus it also lies in $[0,3]$, and the constrained admissibility requirement does not change the projection optimum. Dividing by six gives $1/5400$.

The original v is the displayed affine constant plus a single ReLU, and has zero error. Its native hinge, two-triangle piecewise-affine encoding, and dictionary containing that hinge are exactly the same function, so they induce the same policy

at every state and price. No representational renaming changes either conclusion. *Q.E.D.*

For the worst-state conditional criterion, (34) gives $\sup_x \mathbb{E}_p[J^{b_h} - J^*] = \|h - v\|_\infty^2/6$. The sharp attained approximation errors in Proposition A.10 therefore yield $1/96$ and $1/1536$. The explicit attaining additive and quadratic functions shifted by one stay inside $[0,3]$. The population and worst-state optimization problems are distinct, so their minimizers need not coincide. The exact purchase equation follows by subtracting the respective policy and complete computation costs and solving the strict linear inequality. *Q.E.D.*

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